

3 **B** $a = (v - u)/t = (0 - 15)/1.2 = -12.5 \text{ m s}^{-2}$

$$v^2 = u^2 + 2as$$

$$0^2 = 15^2 + 2(-12.5)s$$

$$s = 9.0 \text{ m}$$

$$\text{Closest distance} = 9.0 + 15(0.10) = 10.5 \text{ m}$$

4 **D** For a (perfectly) elastic collision between two bodies the relative speed of